

2023

Time : 3 hours

Full Marks : 35

Pass Marks : 16

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer any five questions.

1. What do you mean by capacitor ? Express its color code. 5 + 2 = 7
2. Discuss the theory of transformer with construction. Write the different types of loss of it. 5 + 2 = 7
3. Explain Lissajous Figures. 7
4. Draw characteristic curve of a Zener diode. Discuss its use in power supply unit as a regulator. 5 + 2 = 7

5. Discuss P-N junction diode as a full-wave rectifier. 7
6. Discuss radio wave transmission and reception with the help of block diagram. 7
7. What is an antenna ? Discuss how power is radiated out by a dipole antenna ? $5 + 2 = 7$
8. How mobile communication system work ? 7
9. Write notes on any two of the following : $3\frac{1}{2} + 3\frac{1}{2} = 7$
- (a) Color code of carbon resistance
 - (b) CRO
 - (c) Star connection of resistance
 - (d) Transistor

